SIMATS ENGINEERING

ASSESSMENT TOOL 3 TECHNICAL PRESENTATION

COURSE NAME: Cryptography and Network Security
COURSE CODE: CSA5103

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 10%

QUESTIONS: 5 **Total Marks:100**

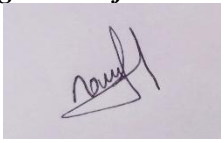
Annexure A

TECHNICAL PRESENTATION

Code of Conduct:

I Naveen G / 192411019 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Annexure A

TECHNICAL PRESENTATION

1. From Caesar to Vigenère: Evolution of Substitution Ciphers”
2. “Substitution vs Transposition: Which Provides Better Security?”

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding g; limited or partial integration	Poor understanding g; unable to integrate concepts
Experiment al Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation on with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation



Caesar vs Vigenère

Substitution vs Transposition

COURSE CODE: CSA5103
COURSE NAME: Cryptography and Network Security

PRESENTED BY:

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SUPERVISOR:

Dharani R

What is a Substitution Cipher?

A substitution cipher replaces each plaintext symbol with another symbol according to a rule or key.

PLAINTEXT

HELLO

Readable message

→

RULE / KEY

SHIFT +3

Transformation

→

CIPHERTEXT

KHOOR

Encrypted message

PLAINTEXT

Original readable information.

KEY

Controls the substitution.

CIPHERTEXT

Protected-looking output.

Caesar Cipher

A fixed-shift substitution: every letter moves by the same number of positions.

ABCDEFGHIJKLMNOPQRSTUVWXYZ

↓ ↓ ↓ ↓ ↓ ↓ ↓ ↓ ↓ ↓ ↓ ↓ ↓ ↓

DEFGHIJKLMNOPQRSTUVWXYZABC

EXAMPLE

ATTACK → DWWDFN

Key = 3

FORMULA

C = (P + K) mod 26

P = (C − K) mod 26

LIMITATION

Only 26 shifts exist.

Brute-force attacks are easy.

Vigenère Cipher

A keyword selects a different Caesar shift for each position.

PLAINTEXT **ATTACKATDAWN**

KEY **LEMONLEMONLE**

CIPHERTEXT **LXFOPVEFRNHR**

ENCRYPTION

$C_i = (P_i + K_i) \bmod 26$

DECRYPTION

$P_i = (C_i - K_i) \bmod 26$

WHY BETTER?

The same plaintext letter can produce different ciphertext letters.

How Substitution Ciphers Work

The plaintext characters are replaced according to a fixed or changing mapping.

PLAINTEXT

A B C D E F

↓ ↓ ↓ ↓ ↓ ↓

D E F G H I

Example mapping: Caesar shift +3

EXAMPLE

ATTACK → DWWDFN

STRENGTH

A complex mapping can provide a large key space.

WEAKNESS

Frequency patterns can reveal repeated symbols.

How Transposition Ciphers Work

The characters stay the same, but their positions are rearranged using a route or key.

PLAINTEXT GRID

W	E	A
R	T	H
E	R	E

READ BY A DIFFERENT ROUTE

EXAMPLE

Rail Fence and Columnar Transposition rearrange character positions.

STRENGTH

Positions are disrupted while the original symbols remain.

WEAKNESS

Letter frequencies remain visible to an attacker.

Caesar vs Vigenère

Feature	Caesar	Monoalphabetic	Vigenère
Substitution	Fixed shift	Random mapping	Multiple shifts
Key	Number	Alphabet permutation	Keyword
Alphabets used	1	1	Multiple
Frequency analysis	Very easy	Possible	Much harder
Security	Very low	Low–moderate	Higher

EVOLUTION: SINGLE SHIFT → RANDOM SUBSTITUTION → CHANGING SHIFTS

Substitution vs Transposition

FEATURE	SUBSTITUTION	TRANSPOSITION
What changes?	Symbol identity	Symbol position
Plaintext letters	Replaced	Preserved
Main clue	Frequency patterns	Position patterns
Examples	Caesar, Vigenère	Rail Fence, Columnar
Typical weakness	Frequency analysis	Structural analysis